Introduction to Artificial Intelligence

Course Code: CE105

Department: Computer Engineering (CE)

- **Course Objectives:**

- Understand the fundamentals of Artificial Intelligence.
- Learn about intelligent agents and problem-solving techniques.
- Understand basic Machine Learning concepts.
- Develop analytical and critical thinking skills.
- Apply AI concepts to solve academic and real-world problems.

- **Course Outcomes:**

After completing this course, students will be able to:

- Explain the basic concepts of Artificial Intelligence.
- Design simple AI-based solutions.
- Apply search algorithms to solve problems.
- Understand the basics of Machine Learning.
- Use AI concepts in academic assistance systems.

- **Course Modules:**

Unit 1: Introduction to Artificial Intelligence

- Definition and History of AI
- Applications of AI
- Intelligent Agents

Unit 2: Problem Solving

- State Space Search
- Breadth First Search
- Depth First Search
- Heuristic Search

Unit 3: Machine Learning Basics

- Supervised Learning
- Unsupervised Learning
- Classification
- Regression

Unit 4: Knowledge Representation

- Logic
- Semantic Networks
- Frames
- Rule-Based Systems

Unit 5: AI Applications

- Chatbots
- Recommendation Systems
- Academic Assistance
- Healthcare
- Education

- **Assessment Pattern:**
- Internal Assessment: 30 Marks
- End Semester Examination: 70 Marks

- **Reference Books:**
- Artificial Intelligence – Stuart Russell & Peter Norvig
- Artificial Intelligence: A Modern Approach
- Introduction to Machine Learning